



Lessons

- Table of Contents
- Confidence Levels
- Notes
- Bookmarks
- Highlights

Solving by Factoring

< >

Solving by Factoring



Solving by Factoring

When you factor an expression, you rewrite it as a product.

$ab + ac \leftarrow$ This expression is the sum of two terms.

$a(b + c) \leftarrow$ The factored expression is the product of a and the quantity $(b + c)$.

Factoring makes it possible to solve quadratic equations because of the zero property of multiplication. It states that when the product of two numbers is 0, at least one of the numbers must be 0. For example, if $ab = 0$, then either $a = 0$, $b = 0$, or both equal 0.

Example 1

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Solve for x .

$$x^2 - 2x = 15$$

Step 1

Set the quadratic expression equal to 0 by subtracting 15 from both sides.

$$x^2 - 2x = 15$$

$$x^2 - 2x - 15 = 0$$

Step 2

Factor the left side of the equation.

$$x^2 - 2x - 15 = 0$$

$$(x + 3)(x - 5) = 0$$

Step 3

The quadratic equation has two solutions: $\{-3, 5\}$. Either will make the original equation true.

$$x^2 - 2x = 15$$

$$(-3)^2 - 2(-3) = 15$$

$$9 + 6 = 15$$

$$15 = 15 \quad \text{True}$$

$$x^2 - 2x = 15$$

$$5^2 - 2(5) = 15$$

$$25 - 10 = 15$$

$$15 = 15 \quad \text{True}$$

The two factors are $(x + 3)$ and $(x - 5)$. The equation is true if either factor equals 0. Set the factors equal to 0, and solve for x .

$$x + 3 = 0$$

$$x = -3$$

$$x - 5 = 0$$

$$x = 5$$

Some equations have only one solution.

Example 2

Hide

Solve for m .

$$m^2 - 8m + 16 = 0$$

The equation is already set equal to 0.

Factor the left side:

$$m^2 - 8m + 16 = 0$$

$$(m - 4)(m - 4) = 0$$

Both factors are the same, so there is only one solution.

$$m - 4 = 0$$

$$\underline{m = 4}$$

The solution set is $\{4\}$.

$$\text{Check: } m^2 - 8m + 16 = 0$$

$$4^2 - 8(4) + 16 = 0$$

$$16 - 32 + 16 = 0$$

$$0 = 0 \quad \text{True}$$

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Lesson 3: Quadratic Equations

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